

Hydrogenic Bound States from Phase Holonomy, Part II

Phase Winding and the Conditional Coulomb / Bohr Baseline

Author: Micah Blumberg

Self Aware Networks Research Institute

<https://selfawarenetworks.com>

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Abstract

Hydrogen's discrete bound states are ordinarily described by quantum mechanics using an electromagnetic Coulomb interaction. Any proposed alternative must derive the source, interaction, stable states, and observable spectrum without quietly inserting those targets as assumptions. The published source named this alternative framework Super Information Theory (SIT) and claimed that it derives Coulomb attraction, stable hydrogenic shells, and their discreteness from a time-density field and phase holonomy [1]. This correction finds that the displayed argument does not establish those claims. The exterior profile $\rho_t(r) = \rho_0(1 - \kappa/r)$ is an assumed harmonic ansatz until an action, source, boundary data, and matched solution are supplied. The expression called an energy density is dimensionally inconsistent with the source's stated units for its coefficient: one reading yields a force density, while the other yields an energy gradient that is not shown to be the interaction potential of an electron-like excitation. Phase degree, connection holonomy, and gauge-covariant circulation are distinct objects. Moreover, $\pi_1(R^3 - \{0\}) = 0$, so an ordinary nonvanishing complex scalar does not acquire protected orbital winding merely by removing a point proton. A line-defect family demonstrates that even where integer winding exists, it does not select a radial scale. A bounded consistency result remains: after assuming $V(r) = -K/r$, circular relative-coordinate dynamics, and the old condition $L = N\hbar$, the Bohr N^2 radius and $1/N^2$ energy scalings follow, with the electron-proton reduced mass. This is not a Schrodinger, Dirac, or QED spectrum. The paper's candidate endpoint is therefore a conditional Bohr reconstruction joined to obstruction and boundary statements about the displayed source arguments, not a general no-go theorem and not a first-principles derivation of hydrogen from SIT.

Scope and evidence boundary

A Corrective Boundary Result for Hydrogenic Claims in Super Information Theory

Source lineage: proposed successor to Zenodo record 18042075, concept DOI 10.5281/zenodo.18042074

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1. Corrective Purpose and Result Classes

In established hydrogen physics, a specified Coulomb Hamiltonian and its boundary conditions lead to a tested spectrum. The scientific problem here is whether a different field model can generate the interaction, stable states, and measurements rather than assume them. The source calls its proposed field framework Super Information Theory; this paper tests the displayed causal steps and records exactly where they succeed, remain assumptions, or fail.

The immutable source record is the object being corrected, not evidence for the revised claims [1]. This manuscript separates four result classes that the source combined:

1. **Derived here:** elementary dimensional consequences, the exterior form of a spherical harmonic function under stated hypotheses, the topology of a point-deleted three-dimensional domain, a kinematic fixed-winding family with continuous scale, and conditional Coulomb / Bohr algebra.
2. **Assumed:** the exterior Laplace model, its coefficient and sign, any test-body response, the Coulomb coefficient, circular motion, the old angular-momentum condition, and all gauge data.
3. **Open:** the SIT action, proton source, matched localized solution, coupling to an electron-like excitation, existence and stability, self-adjoint spectrum, and precision observable map.
4. **Retracted:** first-principles Coulomb recovery, topological necessity of proton-centered shells, stability from positivity alone, standard-spectrum recovery, spectroscopy predictions, and macroscopic gravitational predictions.

This classification changes the endpoint. A corrective paper need not invent the missing action or spectrum to state a bounded correction. It must instead show exactly which implications fail, retain only results that follow from declared assumptions, and leave the stronger program visibly open. That is the endpoint pursued here.

1.1 Preserved first-principles research question

The correction does not reduce the larger research question to the historical Bohr calculation. The distinctive open question is whether one complete, dimensionally consistent SIT model can generate a hydrogenic bound system without inserting the target Coulomb interaction or conventional spectrum as an assumption. A successful result would have to join source dynamics, matter response, gauge structure, existence, stability, and spectral prediction in one controlled derivation.

That program requires at least six linked achievements. First, a declared action must produce the field equations and conservation laws. Second, a localized proton-like source with stated interior and exterior boundary conditions must exist. Third, a specified electron-like sector must couple to that source with the correct sign and normalization rather than by identifying an arbitrary background gradient with force. Fourth, the joined configuration must satisfy an existence and stability analysis that includes constraints, zero modes, radiation, and allowed perturbations. Fifth, a self-adjoint radial operator or controlled nonlinear spectral problem must produce converged states and observables. Sixth, parameters and uncertainties must be mapped to out-of-sample measurements without using those measurements to define the claimed prediction.

None of those requirements is logically ruled out by the obstruction results in this paper. The point-defect topology result rules out one displayed scalar winding argument; the continuous-scale family rules out radius selection by winding alone; and the dimensional audit rules out one unqualified force identification. They do not prove that no future SIT action could contain a different source, bundle, coupling, or spectral mechanism. This distinction is why the first-principles program remains outside the correction rather than being silently abandoned or represented as completed.

2. The Missing Dynamical Bridge

2.1 Conditional exterior profile

The source calls ρ_t a local time-density or causal-update rate, but it supplies no measurement operation that turns that name into an observable. This paper therefore treats ρ_t only as the displayed scalar function; a physical interpretation would have to be earned by a source equation, boundary solution, and measurement map.

The source uses

$$\rho_t(r) = \rho_0 \left(1 - \frac{\kappa}{r}\right), \quad r > r_{\text{core}}. \quad (1)$$

This has the correct units if $[\rho_t] = T^{-1}$ and $[\kappa] = L$. Units do not establish a field equation or a source.

Proposition 1 (conditional harmonic exterior form). Let $\Omega = \{x \in \mathbb{R}^3 : |x| > r_{\text{core}}\}$. Suppose $\rho_t \in C^2(\Omega)$ is static, exactly spherically symmetric, harmonic on Ω , and has a finite limit at spatial infinity. Then

$$\rho_t(r) = A + \frac{B}{r}. \quad (2)$$

Derivation. Spherical symmetry reduces $\text{laplacian}(\rho_t) = 0$ to

$$\frac{d}{dr} \left(r^2 \frac{d\rho_t}{dr} \right) = 0. \quad (3)$$

Integrating twice gives the stated form. The asymptotic condition sets A; source and matching data determine B. Writing $A = \rho_0 > 0$ and $B = -\rho_0 \kappa$ is an additional parameterization. If ρ_t is a positive rate, the physical exterior branch must also satisfy $\rho_0 (1 - \kappa/r) > 0$ throughout Ω ; for positive κ , a sufficient condition is $r_{\text{core}} > \kappa$.

The proposition is exact under these hypotheses, while the source uses the profile as an approximate ansatz. It is conditional in every scientifically important respect. It does not show that the SIT field equation is Laplace's equation, that a proton is its source, that the interior is regular, that matching is possible, or that κ has the required sign and normalization. A mass term could give a Yukawa tail, and nonlinear or coupled equations could give other behavior. The source therefore supplies an exterior ansatz, not a proton solution.

2.2 Energy versus energy-density fork

The source denotes an unspecified dimensionless coherence factor by R_{coh} . Because no operational definition or response law is supplied, the audit treats it as an arbitrary dimensionless function and asks whether its spatial variation changes the claimed force.

The source writes

$$E(x) = \zeta R_{\text{coh}}(x) \rho_t(x)^2. \quad (4)$$

while calling E an energy density and assigning

$$[R_{\text{coh}}] = 1, \quad [\rho_t] = T^{-1}, \quad [\zeta] = ML^2. \quad (5)$$

Proposition 2 (dimensional fork). Those statements cannot all hold.

With the stated coefficient units,

$$[\zeta R_{\text{coh}} \rho_t^2] = ML^2 T^{-2} = J. \quad (6)$$

so the expression is an energy. If the density label is retained instead, then $[\text{zeta}] = M L^{-1}$, and

$$[-\nabla E] = M L^{-2} T^{-2} = N m^{-3}. \quad (7)$$

which is a force density. Under neither reading does dimensional consistency identify the expression as an electron's interaction potential. The source also defines $m = E/c^2$. On the stated-zeta branch this has mass units but lacks an ontology specifying whether it is a point, cell, or distributed quantity. On the density branch it is a mass density and requires a spatial measure and integration rule before a total mass exists.

Under the density reading, a force on a localized extended excitation would require an interaction functional such as

$$U[X] = \int W(x - X) \delta \varepsilon_{bg}(x) d^3x, \quad F[X] = -\nabla_X U[X]. \quad (8)$$

with a specified response W , support, normalization, and backreaction. Under the stated-zeta reading, one could postulate $U(X) = g E_{bg}(X)$, but the response coefficient and its sign would be new model data. The source gives neither coupling.

For the full radial source expression,

$$E(r) = \zeta \rho_0^2 R_{coh}(r) \left(1 - \frac{\kappa}{r}\right)^2. \quad (9)$$

the derivative is

$$-\frac{dE}{dr} = -\zeta \rho_0^2 \left[R'_{coh}(r) \left(1 - \frac{\kappa}{r}\right)^2 + \frac{2\kappa R_{coh}(r)}{r^2} \left(1 - \frac{\kappa}{r}\right) \right]. \quad (10)$$

The first term can change the sign or radial power and cannot be hidden in a constant. A leading inverse-square term follows only on an explicitly controlled branch, for example when $R_{coh}(r) \rightarrow R_{infinity} \neq 0$ and $R'_{coh}(r) = 0$ (κ/r^2), together with a position-independent response coefficient. In the narrower constant- R_{coh} case, write

$$Q(r) = C \left(1 - \frac{\kappa}{r}\right)^2. \quad (11)$$

one obtains

$$-\frac{dQ}{dr} = -\frac{2C\kappa}{r^2} + \frac{2C\kappa^2}{r^3}. \quad (12)$$

The leading inverse-square gradient is then valid for $r \gg |\kappa|$. Calling it a universal attractive particle force still requires the physical branch, response law, sign, and coefficient. In particular, this calculation does not derive

$$K = \frac{e^2}{4\pi\varepsilon_0}. \quad (13)$$

opposite responses for opposite charges, or Gauss-law electromagnetism.

3. Phase, Gauge Structure, and Topology

3.1 Three objects, not one integer

A phase can return around a closed path in three distinct senses: the scalar phase may wind, a gauge connection may accumulate a loop phase, and their gauge-covariant combination may circulate. Calling all three effects “phase holonomy” hides different inputs and different failure conditions, so the paper separates them before naming the mathematical objects.

Let a complex scalar be represented in a local trivialization as

$$\psi = Re^{i\theta}. \quad (14)$$

and let γ be a closed loop on which R is nonzero. The phase map may have degree

$$\oint_{\gamma} \nabla \theta \cdot d\ell = 2\pi w, \quad w \in \mathbb{Z}. \quad (15)$$

Choose the convention

$$D = \nabla - i\frac{q}{\hbar}A. \quad (16)$$

Under a gauge change with scalar parameter χ ,

$$\psi' = e^{i(q/\hbar)\chi}\psi, \quad \theta' = \theta + \frac{q}{\hbar}\chi, \quad A' = A + \nabla\chi. \quad (17)$$

Thus $\text{grad}(\theta) - (q/\hbar)A$ is invariant. The integer assigned to a local phase representative can shift under a large change of trivialization, while the connection term shifts with it; the combined circulation is the invariant object.

Then the relevant quantities are:

$$\begin{aligned} w &= \frac{1}{2\pi} \oint_{\gamma} \nabla \theta \cdot d\ell, \\ W_A &= \exp \left[i\frac{q}{\hbar} \oint_{\gamma} A \cdot d\ell \right], \\ C_{\gamma} &= 2\pi w - \frac{q}{\hbar} \Phi_{\gamma}, \quad \Phi_{\gamma} = \oint_{\gamma} A \cdot d\ell. \end{aligned} \quad (18)$$

Only the first expression is integer-valued by phase degree. The Wilson loop is a connection observable, as emphasized by the Aharonov-Bohm effect [10]. The covariant circulation is gauge invariant under the stated convention but is not generally 2π times an integer. Integer covariant circulation requires

$$\frac{q}{\hbar} \Phi_{\gamma} \in 2\pi\mathbb{Z}. \quad (19)$$

equivalently a trivial Abelian Wilson loop in the stated convention. No fluxoid relation is claimed because the source supplies no condensate dynamics. A contractible loop can still have nontrivial connection holonomy when curvature threads a spanning surface; that holonomy need not be topological or quantized.

On a nontrivial bundle, one global theta and one global potential need not exist. Patchwise potentials and transition functions replace the global formula; the magnetic-monopole construction is the standard comparison [7, 8]. Neither a monopole bundle nor a flux quantization law is specified in the hydrogen source.

3.2 A point proton does not create orbital winding

Proposition 3 (point-defect boundary). For an ordinary complex scalar that is nonzero on $\mathbb{R}^3 \setminus \{0\}$, a loop around the removed point carries no protected nonzero scalar phase winding.

Derivation. The point-deleted domain deformation retracts to S^2 , and

$$\pi_1(\mathbb{R}^3 \setminus \{0\}) = \pi_1(S^2) = 0. \quad (20)$$

Every loop contracts. A nonvanishing scalar phase map along that loop extends through the contraction, so its degree must be zero.

This statement concerns scalar loop winding on the base. It does not say that all $U(1)$ bundles over the surrounding sphere are trivial. Nonzero first Chern class is different additional structure and describes magnetic flux, not an automatic electric-proton orbital sector [7, 8].

3.3 Winding does not select radius

A line or vortex defect can support noncontractible loops. That does not make the winding number a radial quantum number.

Proposition 4 (fixed-winding continuous-scale counterexample). In cylindrical coordinates about the z axis, define

$$\begin{aligned} s &= \sqrt{x^2 + y^2}, & \varphi &= \text{atan2}(y, x), \\ \psi_{a,w}(s, \varphi) &= \left(\frac{s}{\sqrt{s^2 + a^2}} \right)^{|w|} e^{iw\varphi}, \\ a &> 0, & w &\in \mathbb{Z} \setminus \{0\}. \end{aligned} \quad (21)$$

For every positive a , the field has phase degree w around the defect axis, while its core scale changes continuously with a . Thus winding alone cannot select a radius.

For $w=0$, the field is identically one and has no vortex core; it is the trivial sector and is not part of the scale-varying counterexample.

The family is a kinematic counterexample, not an SIT solution or a claim of finite total energy. A global vortex may have an infrared divergence in an unbounded domain. Gauge screening, boundary conditions, and dynamics may select a scale only after they are specified. Nielsen and Olesen provide a primary comparator for the additional field content and equations a local vortex requires [12].

4. The Bounded Positive Result

4.1 Reduced-mass Coulomb mechanics

Let

$$\mu = \frac{m_e m_p}{m_e + m_p}. \quad (22)$$

be the electron-proton reduced mass. Assume the relative-coordinate potential

$$V(r) = -\frac{K}{r}, \quad K > 0. \quad (23)$$

and a conserved classical angular momentum L . The effective radial potential is

$$V_{\text{eff}}(r) = \frac{L^2}{2\mu r^2} - \frac{K}{r}. \quad (24)$$

Its stationary circular separation and curvature are

$$r_* = \frac{L^2}{\mu K}, \quad \left. \frac{d^2 V_{\text{eff}}}{dr^2} \right|_{r_*} = \frac{K}{r_*^3} > 0. \quad (25)$$

This is a local radial minimum at fixed L in the assumed conservative point-particle problem. It is not a proof of radiative, angular, field-theoretic, gauge, or nonlinear stability.

4.2 Conditional Bohr scaling

Now add the old quantum condition

$$L = N\hbar, \quad N = 1, 2, 3, \dots \quad (26)$$

to obtain

$$r_N = \frac{N^2 \hbar^2}{\mu K}, \quad E_N = -\frac{\mu K^2}{2\hbar^2 N^2}. \quad (27)$$

These are the historical Bohr scalings [3]. The separation for $N=1$ uses μ ; it is slightly larger than the infinite-proton-mass radius $a_0 = \hbar^2 / (m_e K)$. The electron and proton distances from the center of mass are different mass-weighted fractions of r_N .

Proposition 5 (conditional Bohr reconstruction). Assumed Coulomb dynamics, circular relative-coordinate motion, and $L=N\hbar$ imply the displayed N^2 radius and $1/N^2$ energy scalings and a local radial minimum at fixed L .

This bounded algebra is not a Schrodinger, Dirac, QED, or SIT spectrum.

The proposition is exact within its assumptions but does not derive any of them from SIT. General semiclassical quantization requires action conditions for the independent cycles and appropriate turning-point corrections; it is not supplied by a single orbital phase statement [9].

5. Why This Is Not the Hydrogen Spectrum

Schrodinger's 1926 paper supplies the historical eigenvalue formulation [4]. A modern nonrelativistic Coulomb spectrum requires a self-adjoint Hamiltonian, a specified operator domain and boundary conditions, and normalizable eigenfunctions; Kato supplies the relevant operator-theory anchor [5]. Relativistic structure requires a separately controlled Dirac-Coulomb problem; Dirac's 1928 paper is historical context rather than a modern singular-operator domain result [6].

The old Bohr integer is not the modern orbital quantum number. For principal number N , standard hydrogen permits

$$\ell = 0, 1, \dots, N - 1, \quad L^2 = \ell(\ell + 1)\hbar^2. \quad (28)$$

The old assignment $L=N\hbar$ is not this angular-momentum spectrum. The source also gives no SIT radial operator, operator domain, boundary conditions, normalizability criterion, degeneracy calculation, transition matrix element, or line-width model. Its conditional energy formula therefore does not recover wavefunctions, degeneracies, fine structure, hyperfine structure, Lamb shifts, transition amplitudes, selection rules, or widths.

6. Spectroscopy as a Boundary Check

Using nominal constants from the 2022 CODATA global adjustment [14], the bounded checker evaluates the old Bohr principal-index gap as

$$f_{N=1 \rightarrow N=2}^{\text{conditional Bohr gross}} \approx 2.46604 \times 10^{15} \text{ Hz}. \quad (29)$$

Parthey et al. measured the hydrogen 1S-2S hyperfine-centroid frequency as [13]:

$$f_{1S-2S}^{\text{measured}} = 2,466,061,413,187,035(10) \text{ Hz}. \quad (30)$$

The nominal arithmetic difference is approximately $2.30\text{e}10$ Hz. It has no theory uncertainty and is not a significance test, SIT residual, exclusion, anomaly, or new-physics bound. The old Bohr model has only the principal index N ; it does not derive 1S and 2S state labels. Its use of the electron-proton reduced mass already includes the leading finite-mass correction, while relativistic, radiative / QED, higher-order recoil and radiative-recoil, nuclear-size, and polarizability contributions are absent. Because the Parthey datum is input A6 to the same CODATA adjustment used for the constants, the two sides are not statistically independent without a covariance and adjustment analysis. The comparison is included only to prevent nominal gross-gap arithmetic from being mislabeled as precision spectroscopy.

A future spectroscopy claim requires state-resolved energies and wavefunctions, an observable-specific term inventory through a declared perturbative order, a parameter-to-observable map, transition and apparatus definitions, numerical convergence, fitting rules, uncertainty propagation, and out-of-sample predictions. None is supplied here.

7. Cross-Paper Consistency: Part I Is Lineage, Not Support

Part I is Zenodo record 18042095, DOI 10.5281/zenodo.18042095, and concept DOI 10.5281/zenodo.18011429 [2]. It cannot be cited as independent validation of Part II for four reasons.

First, Part I says that Part II establishes the necessity of discrete hydrogenic structure. That statement is superseded by the present correction: Part II does not establish necessity.

Second, Part I writes the covariant-circulation expression

$$\oint \left[\nabla\theta - \frac{q}{\hbar}A \right] \cdot d\ell = 2\pi n. \quad (31)$$

without displaying the additional flux condition needed for that equality. The distinction among phase degree, Wilson holonomy, and covariant circulation in Section 3 applies to both papers.

Third, Part I's Coulomb and spectral results are embeddings of assumed Maxwell / Schrodinger / Dirac dynamics in a declared QFT limit. They do not supply the full SIT action, prove that the limit follows from it, derive the proton source of Part II, or construct an SIT hydrogen spectrum. Standard physics can be recovered after being placed in an effective action; that does not show that SIT independently derives it.

Fourth, Part I repeats the swapped record-title pair found in Part II. Record 17270488 is the electron-holonomy work, while record 17311157 is the Yang-Mills work. Their concept-DOI lineages are cited here only to preserve the research genealogy [15, 16].

Part I remains a separate revision target. This manuscript does not revise its public record, infer that all of its equations are invalid, or use its unreviewed SIT-specific assertions as evidence. Until the cross-paper corrections are aligned, neither paper may validate the other.

8. Stability and Existence Remain Open

Positivity of a term such as

$$\int R^2 \left| \nabla\theta - \frac{q}{\hbar}A \right|^2 d^3x. \quad (32)$$

means that the term penalizes covariant phase gradients. It does not prove that the complete energy is bounded in the proposed sector, that a localized stationary solution exists, or that the solution is dynamically stable.

A defensible stability result requires a complete action and constraints, a finite-energy stationary solution of all equations, treatment of gauge and other zero modes, a constrained second variation or stronger nonlinear argument, and control of radiation and time-dependent perturbations. Derrick's scaling result is a required comparator for ordinary static scalar lumps [11]. It is not a universal prohibition: gauge fields, conserved charges, topology, time dependence, or higher derivatives may alter the conclusion when they are present in the declared action.

9. Limitations and Reproducible Bounded Checks

model/check_conditional_hydrogen_model.py uses only the Python standard library. It checks:

1. the energy / mass versus density fork;
2. the full $R_{\text{coh}}(r)$ product derivative and its controlled constant or asymptotic branch;
3. fixed nonzero winding with continuously variable line-defect core scale and the separate $w=0$ sector;
4. gauge-transformation and flux dependence of covariant circulation;
5. conditional reduced-mass Coulomb / Bohr radii and energies;

6. the mismatch between the Bohr integer and modern orbital numbers;
7. the scope difference between gross structure and measured spectroscopy;
8. Draft 5 artifact, citation-key, claim-ledger, authorization, and no-typeset guards.

An overall: PASS result means only that these bounded checks and package guards replayed. The script does not prove the analytical topology statements, solve an SIT field equation, construct a proton, establish stability, or solve a Schrodinger, Dirac, QED, or SIT spectrum.

10. Open First-Principles Program

A stronger SIT hydrogen derivation requires all of the following:

1. one dimensionally consistent action and convention ledger;
2. a proton / source sector that does not insert the target Coulomb field;
3. derived field equations, conserved quantities, stress-energy, test-body coupling, and backreaction;
4. a converged interior / exterior boundary-value solution;
5. existence and constrained or dynamical stability of localized sectors;
6. a declared scalar and bundle topology compatible with the three-dimensional hydrogenic geometry;
7. a self-adjoint linear operator or controlled nonlinear spectral problem;
8. converged eigenvalues, eigenfunctions, degeneracies, and transitions;
9. recovery of declared Schrodinger, Dirac, and QED limits; and
10. uncertainty-controlled comparison with precision observables.

The missing items are not placeholders to be filled by prose. They are the substantive content a future first-principles paper would have to supply.

10.1 Evidence-decision table

The following ladder fixes what each result in this package can and cannot decide. It prevents successful arithmetic or package replay from being used as a substitute for the missing physics.

Evidence class	Present package evidence	Permitted conclusion	Conclusion still blocked
Source correction	Page-located claim ledger and immutable source hash	The listed public claims are narrowed, replaced, or retracted in this successor	The successor theory is physically correct
Exact conditional mathematics	Harmonic exterior form under stated hypotheses; point-defect topology; fixed-winding continuous-scale family	The displayed implications hold inside their explicit assumptions	The assumed field equation, source, or topology follows from SIT
Conditional mechanics	Reduced-mass Coulomb mechanics plus $L=N\hbar$	Historical Bohr radius and energy scaling follow from those inputs	Coulomb attraction, the old quantum condition, or a modern spectrum is derived
Executable regression	Dimensional, derivative, winding, gauge, Bohr, and package checks	The coded formulas and release guards replay deterministically	An SIT field solution, stability theorem, or atomic spectrum exists
Precision boundary check	Nominal gross-gap arithmetic and an explicit omitted-term inventory	A gross Bohr number must not be mislabeled as precision spectroscopy	An anomaly, exclusion, residual, or new-physics parameter bound
Discriminating physical evidence	None in this package	No positive physical verdict	Any first-principles SIT hydrogen claim

A future result moves into the final row only if it derives a nontrivial observable from a frozen model, propagates uncertainty, survives relevant comparators and nulls, and predicts data not used to construct or tune the result. Agreement after assuming the Coulomb potential is a consistency check, not discriminating evidence for SIT.

11. Endpoint and Claim Discipline

The endpoint of this private final correction candidate is:

Conditional Bohr reconstruction plus source-specific obstruction and boundary results.

Under an assumed Coulomb potential and old circular-orbit condition, the reduced-mass Bohr scaling follows. The published time-density, phase-winding, topology, and positivity arguments do not derive the assumed interaction, select the radius, establish stability, or produce the modern hydrogen spectrum.

This is the bounded endpoint for the displayed source arguments, not a general no-go theorem for every possible SIT model. The exact local final-preprint package is complete only after its manuscript, model, source / render, privacy, and all-page checks pass. Independent mathematical-physics and spectroscopy review, author approval, and repository release remain separate external release gates; they are not evidence that the first-principles SIT program is complete.

No correction notice, Zenodo version, OSF change, GitHub mirror, venue submission, or public repository release is authorized by this draft. A private review PDF may be rendered under the 69-paper quality program; that production artifact does not authorize release. The immutable historical record must remain identifiable, and any later version must retain the concept-DOI lineage only after explicit author approval.

Authorial provenance and research chronology

Artificial intelligence was not involved in authoring or originating the core ideas, scientific theories, hypotheses, or source-based research corrected and developed in this paper. The underlying research program is Micah Blumberg's original work and forms part of a continuous research history spanning 2011-2026, with central conceptual foundations recorded before ChatGPT's public release on November 30, 2022.

During later manuscript refinement, AI tools were used in bounded supporting roles including mathematical and computational formulation, code, grammar and spelling, and checks against physics and mathematical literature. These tools did not supply the original theory or authorship. Micah Blumberg selected the claims and sources, directed the correction, and remains responsible for the paper's arguments, interpretations, and errors.

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